Information-Theoretic Limits of Explainability: When Can We Trust XAI Under Model Compression?

Abstract

Post-hoc explainability methods are increasingly used to justify compressed models in resource-constrained settings, often assuming that preserved predictive performance implies preserved explanation reliability. We show that this assumption can fail fundamentally: model compression introduces limits on explanation recovery even when task performance remains largely unchanged. Building on recent information-theoretic formulations of explainability, we model explanation recovery as a cascade of two channels—a compression channel followed by a query/masking channel. We show that the resulting explanation capacity is constrained by the weaker channel and, using Fano's inequality, derive a lower bound on explanation error determined by information lost during compression. This establishes an explanation-error floor below which post-hoc explainers, including SHAP, LIME, and attribution-based methods, cannot reliably recover feature importance through hyperparameter tuning alone.

We instantiate the compression channel through Eckart–Young–Mirsky SVD truncation and evaluate the resulting bound on SVD-compressed Whisper automatic speech recognition models for isiZulu, Setswana, and Sesotho. Across compression ranks, we compare the theoretically predicted error floor against empirical attribution instability, measured via feature-rank reversals, while tracking word error rate (WER). Strikingly, we identify compression regimes in which WER improves while explanation stability simultaneously deteriorates to near-random levels, revealing cases where compression, rather than the explainer, constitutes the dominant bottleneck for faithful attribution. These findings provide an information-theoretic criterion for identifying compression regimes in which explanations should not be trusted, with practical implications for deploying explainable AI in low-resource speech and other resource-constrained machine-learning applications.

Keywords: Explainable AI, Information Theory, Model Compression, Fano's Inequality, Attribution Stability, Low-Resource Speech Recognition, SVD, Trustworthy AI